**6.**

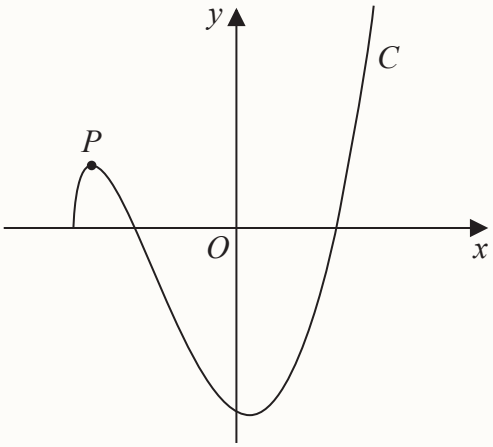



**Figure 3**

**In this question you must show all stages of your working.**

**Solutions relying entirely on calculator technology are not acceptable.**

The function f is defined by

$$f(x) = 5(x^2 - 2)(4x + 9)^{\frac{1}{2}} \qquad x \geqslant -\frac{9}{4}$$

(a) Show that

$$f'(x) = \frac{k(5x^2 + 9x - 2)}{(4x + 9)^{\frac{1}{2}}}$$

where $k$ is an integer to be found.

**(4)**

(b) Hence, find the values of $x$ for which $f'(x) = 0$

**(1)**

Figure 3 shows a sketch of the curve $C$ with equation $y = f(x)$.

The curve has a local maximum at the point $P$

(c) Find the exact coordinates of $P$

**(2)**

The function g is defined by

$$g(x) = 2f(x) + 4 \qquad -\frac{9}{4} \leqslant x \leqslant 0$$

(d) Find the range of g

**(3)**